Part XI Dermatology
Chapter 11C: Viral, Fungal, Infestation, and Benign/Premalignant Skin Disorders

Medvale Internal Medicine - original synthesis for study and Graph-RAG extraction

Chapter orientation: This chapter covers common outpatient and inpatient dermatology problems caused by viruses, superficial fungi, mites, and lice, then transitions into benign and premalignant keratotic lesions. The goal is not only recognition, but also the decision logic: when to test with KOH, mineral oil scraping, viral PCR, culture, or biopsy; when empiric therapy is reasonable; and when urgent dermatology, ophthalmology, or infection control involvement is needed.

1. Diagnostic Framework for Infectious, Infestation, and Keratotic Lesions

Dermatologic diagnosis begins with lesion morphology, distribution, timing, symptoms, and host context. Viral lesions often appear as grouped vesicles, dermatomal vesicles, verrucous papules, or umbilicated papules. Dermatophyte infections usually have scale, annularity, maceration, alopecia, or nail dystrophy. Infestations produce intense pruritus, characteristic body sites, and often household or institutional clustering. Benign keratinocyte lesions tend to be stable and recognizable by texture, whereas premalignant or malignant lesions more often grow, bleed, ulcerate, become indurated, or fail to heal.

A practical first branch point is whether the eruption is contagious. Warts, molluscum contagiosum, dermatophyte infections, scabies, and lice can spread through direct contact or shared objects. Herpes zoster is not transmitted as shingles, but active vesicular fluid can transmit varicella-zoster virus to a susceptible contact and cause primary varicella. Tinea versicolor is caused by overgrowth of normal Malassezia flora and is not managed like a highly contagious household outbreak. Actinic keratosis and seborrheic keratosis are not infectious; their importance is cancer risk stratification and reassurance.

Question	Why it matters	High-yield examples
Is the lesion vesicular, verrucous, annular, umbilicated, burrow-like, or keratotic?	Morphology narrows the differential before testing.	Grouped painful vesicles suggest HSV; unilateral dermatomal vesicles suggest zoster; umbilicated papules suggest molluscum; annular scale suggests tinea corporis; burrows suggest scabies; gritty sun-exposed scale suggests actinic keratosis.
Is there scale?	Scale points toward dermatophyte, psoriasis, eczema, seborrheic dermatitis, pityriasis rosea, tinea versicolor, or actinic keratosis.	KOH is high yield when a scaly rash is annular, asymmetric, steroid-worsened, or treatment-resistant.
Is it painful or pruritic?	Pain suggests zoster, HSV, bacterial infection, ischemia, or necrotizing infection; nocturnal pruritus suggests scabies.	Zoster pain may precede rash; scabies itch can persist after successful treatment.
What is the host context?	Immunosuppression changes severity, distribution, contagiousness, and treatment threshold.	Disseminated zoster, extensive molluscum, crusted scabies, deep fungal disease, and nonhealing keratotic lesions require escalation.
Has treatment failed?	Failure may mean wrong diagnosis, nonadherence, reinfection, resistance, or malignancy mimicking inflammation.	A presumed wart, actinic keratosis, or keratoacanthoma that grows, bleeds, ulcerates, or persists after therapy should be biopsied.

Bedside testing: KOH preparation is the highest-yield quick test for suspected dermatophyte infection and tinea versicolor. Mineral oil scraping can identify mites, eggs, or scybala in scabies. Viral PCR is useful for HSV/VZV when morphology is atypical, the patient is immunocompromised, or the site is high risk. Biopsy is the safety valve for unclear, persistent, indurated, ulcerated, bleeding, rapidly growing, or cancer-suspicious lesions.

2. Viral Skin Infections

2.1 Human Papillomavirus Warts

Cutaneous warts are benign epithelial proliferations caused by human papillomavirus. Transmission occurs by skin-to-skin contact and is facilitated by microtrauma, maceration, shaving, nail biting, and immunosuppression. Warts are usually clinical diagnoses. The source segment highlights common warts as the most common type, flat warts on the face or dorsal hands, plantar warts on pressure-bearing soles, and anogenital warts as condyloma acuminata. Genital HPV types 6 and 11 commonly cause condyloma, whereas types 16 and 18 are associated with cervical and other anogenital cancers.

Wart type	Typical location	Appearance	Clinical note	Common management
Verruca vulgaris - common wart	Hands, fingers, knees, elbows, periungual skin	Flesh-colored or white rough hyperkeratotic papule; thrombosed capillaries may look like black dots.	Most common type; may regress spontaneously but can persist or spread.	Salicylic acid, cryotherapy, paring, curettage, cantharidin, immunotherapy, or observation.
Verruca plana - flat wart	Face, chin, dorsal hands, legs	Multiple smooth, flat-topped flesh-colored papules.	Often numerous; shaving can spread lesions linearly.	Observation, topical retinoid, cautious cryotherapy, or dermatology-directed treatment.

Wart type	Typical location	Appearance	Clinical note	Common management
Plantar wart	Sole, pressure surfaces	Rough hyperkeratotic papule/plaque; interrupts skin lines; pain with lateral compression.	Can be confused with callus; paring may reveal punctate capillaries.	Salicylic acid, cryotherapy, offloading, paring; refractory lesions may need procedural therapy.
Condyloma acuminatum	Genitalia, perineum, anus	Soft fleshy papules or cauliflower plaques.	Common viral STI manifestation; evaluate STI risk, pregnancy, immunosuppression, vaccination, and screening needs.	Patient-applied imiquimod/podofilox/sinecatechins or clinician-applied cryotherapy, trichloroacetic acid, excision, electrosurgery.

Treatment should match lesion site, burden, immune status, symptoms, and patient goals. Many nongenital warts resolve without treatment, but therapy is reasonable if lesions are painful, spreading, cosmetically distressing, or functionally limiting. Salicylic acid requires weeks of adherence. Cryotherapy is common but often requires repeat visits and can cause blistering, dyspigmentation, or pain. Genital wart treatment does not replace cervical cancer screening or high-risk HPV management.

Do not miss: A hyperkeratotic lesion presumed to be a wart should be reconsidered if it is solitary, rapidly enlarging, ulcerated, bleeding, painful without pressure, fixed, or present in an older sun-exposed or immunosuppressed patient. Squamous cell carcinoma can mimic a wart.

2.2 Molluscum Contagiosum

Molluscum contagiosum is a poxvirus infection producing small, dome-shaped, pearly or flesh-colored papules with central umbilication. It is common in children, sexually active adults, athletes with close skin contact, and immunocompromised patients. Lesions are often asymptomatic but may itch or become inflamed during immune clearance. Transmission occurs by skin contact, autoinoculation, shared towels, and occasionally fomites. Extensive, giant, facial, or treatment-refractory lesions should prompt consideration of impaired cellular immunity, including advanced HIV or iatrogenic immunosuppression.

Feature	Molluscum pattern
Morphology	2-5 mm smooth dome-shaped papules with central dell/umbilication; pearly, pink, or flesh-colored.
Distribution	Children: trunk, extremities, flexural areas. Adults: genital, lower abdomen, inner thighs when sexually transmitted. Immunosuppressed: face, diffuse, large, or numerous lesions.
Diagnosis	Usually clinical; biopsy rarely needed. Dermoscopy can show central white/yellow lobules and peripheral vessels.
Treatment	Observation is reasonable because many cases are self-limited. Options include curettage, cryotherapy, cantharidin, topical retinoids, or other dermatology-directed therapy.
Counseling	Avoid sharing towels, reduce scratching, cover lesions during contact sports, and consider STI evaluation for adult genital cases.

2.3 Herpes Zoster

Herpes zoster results from reactivation of varicella-zoster virus in sensory ganglia. The classic pattern is a unilateral painful vesicular eruption in a dermatomal distribution. Pain, burning, tingling, or hyperesthesia may precede the rash by days, which can mimic musculoskeletal pain, myocardial ischemia, renal colic, dental pain, or abdominal disease depending on the dermatome. Vesicles usually crust in 7-10 days, but neuropathic pain may persist as postherpetic neuralgia. Zoster is more common with age over 50 and impaired cell-mediated immunity; zoster in a young patient, disseminated lesions, recurrent episodes, or unusually severe disease should trigger review for immunosuppression.

Complication	Clinical clue	Why it matters
Postherpetic neuralgia	Persistent neuropathic pain after rash resolution; risk rises with age and severe acute pain.	Most common chronic complication; treated with gabapentin/pregabalin, TCAs, topical lidocaine/capsaicin, and multimodal pain control.
Herpes zoster ophthalmicus	V1 distribution, forehead/scalp lesions, eye pain, red eye, visual symptoms; nasal tip lesion raises concern.	Urgent ophthalmology; prompt antivirals reduce ocular morbidity and protect vision.
Ramsay Hunt syndrome	Ear vesicles, facial paralysis, otalgia, hearing symptoms, vertigo.	Prompt antiviral therapy and often steroids; neurologic recovery can be incomplete.
Disseminated zoster	Lesions outside primary/adjacent dermatomes or visceral involvement, especially immunocompromised host.	Consider IV acyclovir, airborne/contact precautions, and evaluation for immune compromise.
Bacterial superinfection	Increasing erythema, warmth, purulence, fever after vesicles appear.	Treat secondary impetigo/cellulitis and reassess for abscess.

Antiviral therapy is most effective within 72 hours of rash onset, but treatment is still appropriate after 72 hours if new lesions are appearing, pain is severe, the patient is immunocompromised, or ophthalmic/neurologic complications are present. Oral valacyclovir and famciclovir are easier to dose than acyclovir for immunocompetent outpatients. IV acyclovir is used for disseminated disease, severe immunocompromise, visceral involvement, or serious neurologic/ocular disease. Analgesia matters because acute pain predicts postherpetic neuralgia. Systemic steroids, if used, are adjunctive in selected patients and never replace antivirals.

Transmission distinction: A patient with zoster does not directly give another person shingles. Contact with active vesicular fluid can transmit VZV to a nonimmune person, causing chickenpox. Cover lesions and avoid contact with pregnant nonimmune people, neonates, and immunocompromised patients until lesions crust.

2.4 Herpes Simplex Virus Skin Disease

HSV is a key vesicular eruption mimic. HSV-1 and HSV-2 produce recurrent grouped vesicles on an erythematous base, often preceded by burning or tingling. Important patterns include orolabial herpes, genital herpes, herpetic whitlow, eczema herpeticum, and HSV keratitis. HSV lesions are usually grouped rather than strictly dermatomal, recur in the same region, and may ulcerate. PCR from a fresh vesicle or ulcer base is preferred when confirmation is needed. Eczema herpeticum is a dermatologic emergency: widespread monomorphic punched-out erosions in a patient with atopic dermatitis can disseminate and requires systemic antiviral therapy.

HSV syndrome	Clue	Management emphasis
Orolabial herpes	Painful grouped vesicles at vermilion border; recurrent cold sores.	Episodic oral antivirals if early; topical agents have limited effect.
Genital herpes	Painful vesicles/ulcers, dysuria, tender nodes; primary episode may be systemic.	Antiviral therapy, STI testing, counseling, suppressive therapy if frequent or transmission reduction desired.
Herpetic whitlow	Painful vesicles on finger; healthcare/dental exposure or autoinoculation.	Do not incise and drain; antivirals for early/severe disease.
Eczema herpeticum	Diffuse monomorphic punched-out erosions, fever, pain in atopic dermatitis.	Urgent systemic acyclovir/valacyclovir; evaluate ocular involvement.
HSV keratitis	Eye pain, photophobia, dendritic corneal lesions.	Urgent ophthalmology; avoid topical steroids unless directed by ophthalmology.

3. Fungal Skin Infections

Superficial fungal infections are divided into dermatophyte infections, yeast infections, and Malassezia-related disease. Dermatophytes digest keratin and infect stratum corneum, hair, and nails. The major genera are Trichophyton, Microsporum, and Epidermophyton. Dermatophyte infections are named by body site: tinea corporis, capitis, pedis, cruris, and unguium. KOH preparation shows branching septate hyphae in dermatophytes; tinea versicolor shows short hyphae and spores described as spaghetti and meatballs. A common diagnostic mistake is topical steroid monotherapy, which can create tinea incognito: less scale, more spread, and a misleading eczematous appearance.

Condition	Location / host	Typical morphology	Diagnosis	First-line treatment logic
Tinea corporis	Body/trunk/extremities; all ages	Annular scaly plaque with raised advancing border and central clearing; often pruritic.	KOH from active scaly border.	Topical terbinafine or azole for localized disease; oral therapy if extensive, immunocompromised, follicular, or refractory.
Tinea capitis	Scalp; mainly children	Scaling alopecia, broken hairs, black dots, posterior cervical nodes; kerion is inflammatory boggy plaque.	KOH/culture; Wood lamp may fluoresce with some Microsporum but not most Trichophyton.	Oral antifungal required; adjunct selenium sulfide/ketoconazole shampoo reduces shedding.
Tinea pedis	Feet, interdigital spaces; adolescents/adults	Maceration, fissuring, scaling, moccasin distribution or vesicles.	Clinical or KOH.	Topical terbinafine/azole plus drying and footwear hygiene; treat nail reservoir if recurrent.
Tinea cruris	Groin/inner thighs; usually scrotum spared	Pruritic annular plaques with scale at edge.	Clinical or KOH; candidiasis if scrotum involved with satellites.	Topical antifungal and moisture control; avoid routine steroid-antifungal combinations.
Tinea unguium / onychomycosis	Toenails > fingernails; older adults, diabetes, tinea pedis	Thickened, yellow, crumbly, dystrophic nails with subungual debris.	Confirm with KOH, PAS clipping, or culture before prolonged oral therapy.	Oral terbinafine often most effective for dermatophyte toenail disease; consider liver/drug interactions.

Treatment shortcut: Tinea capitis and onychomycosis generally need oral antifungal therapy. Most localized tinea corporis, pedis, and cruris can start with topical antifungals plus moisture/friction control. Steroid monotherapy worsens dermatophyte infection.

3.1 Dermatophyte Diagnostic Algorithm

When a patient has a scaly eruption, ask whether the border is annular and active, whether lesions are asymmetric, and whether there is central clearing. Scrape the active edge for KOH. If KOH is positive, treat as dermatophyte and look for reservoirs such as tinea pedis or onychomycosis. If KOH is negative but suspicion remains, repeat scraping, send fungal culture, or biopsy with PAS stain. If the eruption is on the scalp, do not rely on topical therapy; use systemic therapy. If the rash worsens with topical steroid, assume tinea incognito until proven otherwise.

Algorithm step	Action
Scaly annular plaque or macerated interdigital rash	Perform KOH from scale at active edge or macerated area.
KOH shows septate hyphae	Treat dermatophyte; choose topical vs oral based on site and extent.
Scalp or nail involvement	Confirm when possible and use oral therapy; topical therapy alone is inadequate for tinea capitis and usually inadequate for established onychomycosis.
Steroid-modified rash	Stop steroid monotherapy, repeat KOH/culture, and treat suspected tinea incognito.
Recurrent groin/foot disease	Treat reservoirs, reduce moisture, change socks/footwear habits, consider diabetes/immunosuppression if severe.

3.2 Tinea Versicolor

Tinea versicolor is a superficial Malassezia overgrowth that produces hypo- or hyperpigmented macules and patches with fine scale, classically on the upper trunk, neck, and proximal arms. It is more prominent in hot, humid environments and may recur. Pigment change may persist after organisms are eradicated, so persistent dyspigmentation is not immediate treatment failure. KOH shows short hyphae and spores. Treatment can be topical selenium sulfide, ketoconazole shampoo, topical azoles, or oral azoles for extensive or recurrent disease. Oral terbinafine is not useful for tinea versicolor because it does not achieve adequate sweat concentrations against Malassezia.

Feature	Tinea versicolor
Organism	Malassezia species, part of normal skin flora.
Morphology	Fine scaly hypo- or hyperpigmented macules/patches; may coalesce; scale may be subtle and induced by scraping.
Distribution	Upper trunk, shoulders, neck, proximal arms; usually not palms/soles.
Diagnosis	KOH: spaghetti-and-meatballs pattern; Wood lamp may show yellow-green/coppery fluorescence but is not required.
Treatment	Topical selenium sulfide or ketoconazole/azole regimens; oral fluconazole/itraconazole for extensive/recurrent cases. Counsel that pigment recovery lags behind mycologic cure.

3.3 Cutaneous Candidiasis and Intertrigo

Candida favors moist intertriginous areas and compromised barriers. It commonly causes beefy red plaques with satellite pustules in skin folds, diaper areas, beneath breasts, abdominal pannus, groin folds, and around ostomies. Unlike tinea cruris, candidiasis often involves the scrotum and has satellite lesions. Risk factors include diabetes, obesity, antibiotic exposure, immunosuppression, incontinence, occlusion, and moisture. Management requires antifungal therapy plus barrier repair: keep folds dry, reduce friction, use absorbent materials, treat hyperglycemia when relevant, and avoid chronic topical steroid use.

4. Infestations

4.1 Scabies

Scabies is infestation by Sarcoptes scabiei. The mite burrows into the stratum corneum, and symptoms reflect delayed hypersensitivity to mites, eggs, and fecal material. Classic scabies causes severe pruritus, often worse at night, with papules, excoriations, and burrows. Characteristic sites are finger webs, wrists, elbows, axillae, periumbilical region, waistline, buttocks, genitalia, and areolae. In infants and older or immunocompromised adults, scalp, face, palms, and soles can be involved. Scabies should be suspected in persistent generalized pruritus, household clustering, institutional outbreaks, or new pruritus after close contact exposure.

Scabies form	Clues	Management
Classic scabies	Severe nocturnal itch, burrows, papules/excoriations in typical sites; contacts often itch.	Permethrin 5% cream left 8-14 hours and repeated in 7-14 days; oral ivermectin 200 mcg/kg repeated in 7-14 days is an alternative in many nonpregnant adults. Treat contacts simultaneously.
Infant/young child scabies	May involve scalp, face, palms, soles; irritability and poor sleep.	Age-specific guidance; include head/neck if involved; pediatric safety matters.
Crusted scabies	Hyperkeratotic crusted plaques, minimal itch possible, enormous mite burden; immunosuppression, neurologic disease, elderly/institutional risk.	Urgent aggressive therapy with oral ivermectin plus repeated topical permethrin, isolation/contact precautions, environmental decontamination, and outbreak control.
Post-scabetic itch	Itch persists weeks after successful mite eradication; no new burrows or spreading lesions.	Symptomatic topical steroids, antihistamines, emollients; avoid endless retreatment unless reinfestation or active mites suspected.

Diagnosis is clinical when classic, but confirmation can be made by scraping a burrow or papule and identifying mites, eggs, or scybala under microscopy. Dermoscopy may show the delta-wing jet sign. Treatment failure is often due to not treating contacts, inadequate body coverage, washing medication off too early, reinfestation from untreated bedding/clothing, or misdiagnosis. Close contacts should be treated at the same time, even if asymptomatic. Wash clothing, towels, and bedding used recently in hot water and dry on high heat, or seal nonwashable items in a bag for several days.

Crusted scabies red flag: Crusted scabies is highly contagious because of the massive mite burden. It can present with thick scale and surprisingly little itch. Think of it in immunocompromised, elderly, neurologically impaired, or institutionalized patients with widespread crusting or outbreak exposure.

4.2 Pediculosis: Lice

Lice infest hair-bearing regions and cause pruritus through hypersensitivity to saliva. Pediculosis capitis affects the scalp, especially children; diagnosis requires identifying live lice or viable nits close to the scalp. Pediculosis corporis lives in clothing seams and is associated with crowding or limited hygiene access; it can transmit Bartonella quintana, Borrelia recurrentis, and Rickettsia prowazekii in some settings. Pediculosis pubis affects pubic hair and should prompt STI risk assessment. Treatment uses permethrin, pyrethrins, spinosad, topical ivermectin, malathion, or oral ivermectin in selected cases, plus nit combing and decontamination. Body lice treatment focuses on laundering or replacing clothing and bedding.

Infestation	Typical site	Key clue	Treatment emphasis
Head lice	Scalp, behind ears, occiput	Nits attached to hair shafts; live lice confirm active infestation.	Pediculicide plus combing; repeat therapy depending on product; treat confirmed active cases.
Body lice	Clothing seams, trunk	Excoriations, pruritus, poor access to clean clothing; lice in seams.	Launder/replace clothing and bedding; topical treatment often unnecessary if hygiene measures possible.
Pubic lice	Pubic hair, sometimes eyelashes	Pruritus, nits/lice; blue-gray macules possible.	Treat patient and sexual contacts; STI testing; ophthalmic management if eyelashes involved.

5. Benign and Premalignant Keratotic Lesions

Keratotic lesions are common in primary care and internal medicine. The central task is to identify lesions that can be safely recognized versus lesions needing biopsy. Benign lesions usually have symmetry, stable history, uniform color, and classic texture. Premalignant or malignant lesions are more likely to be persistent, tender, indurated, ulcerated, bleeding, rapidly growing, or located on chronically sun-exposed skin. Immunosuppressed patients have higher risk for squamous cell carcinoma and more aggressive behavior, so the threshold for biopsy is lower.

5.1 Actinic Keratosis

Actinic keratoses are premalignant keratinocyte lesions caused by chronic ultraviolet exposure. They present as rough, gritty, scaly macules or papules on sun-exposed sites such as scalp, face, ears, dorsal hands, forearms, and lower lip. The lesion may be easier to feel than to see. Individual AKs have low per-lesion risk of progression, but AKs are markers of field cancerization and cumulative UV damage. Lesions that are tender, thick, indurated, ulcerated, bleeding, rapidly enlarging, or refractory to treatment require biopsy to exclude SCC.

Management category	Examples	When used
Lesion-directed therapy	Cryotherapy with liquid nitrogen, curettage/shave removal when diagnostic uncertainty exists.	Few discrete lesions; hypertrophic lesion where histology may be needed.
Field-directed therapy	5-fluorouracil, imiquimod, tirbanibulin, diclofenac, photodynamic therapy.	Multiple AKs or diffuse photodamage where subclinical lesions are likely.
Prevention and surveillance	Sun protection, hats, sunscreen, avoidance of tanning beds, skin exams.	All patients; especially fair skin, outdoor exposure, transplant/immunosuppressed patients, prior keratinocyte carcinoma.
Biopsy	Shave or punch depending on lesion.	Induration, tenderness, bleeding, ulceration, rapid growth, treatment failure, or concern for SCC.

AK vs SCC: Actinic keratosis is usually thin and gritty. Squamous cell carcinoma is more concerning when the lesion becomes thick, tender, indurated, ulcerated, bleeding, or nodular. Hypertrophic AK and early SCC overlap clinically, so biopsy is the safety valve.

5.2 Seborrheic Keratosis

Seborrheic keratoses are extremely common benign epidermal tumors that increase with age and often run in families. They are not premalignant. They classically appear as waxy, verrucous, stuck-on papules or plaques with color ranging from flesh-colored to tan, brown, gray, or black. They can occur anywhere except mucous membranes, palms, and soles. The key clinical value is reassurance, but inflamed, irritated, bleeding, or diagnostically uncertain lesions may be treated or biopsied. A sudden eruption of numerous seborrheic keratoses has been called the Leser-Trelat sign, but its strength as a paraneoplastic sign is debated; assess the overall clinical context.

Seborrheic keratosis clue	Interpretation
Stuck-on waxy/verrucous surface	Classic benign texture. Dermoscopy may show milia-like cysts and comedo-like openings.
Multiple lesions in older adult	Very common normal aging pattern.
Inflamed or traumatized lesion	Can mimic melanoma or SCC; biopsy if diagnosis is uncertain.
Very dark solitary irregular lesion	Do not assume SK if melanoma cannot be excluded. Use dermoscopy or biopsy.
Treatment	No treatment required; cryotherapy, curettage, or shave removal for irritation/cosmesis/diagnosis.

5.3 Keratoacanthoma

Keratoacanthoma is a rapidly growing crateriform epithelial tumor that clinically resembles well-differentiated squamous cell carcinoma. It often appears on sun-exposed skin as a dome-shaped nodule with a central keratin plug and may enlarge over weeks. Although spontaneous regression can occur, modern practical management treats keratoacanthoma-like lesions with biopsy or excision because clinical distinction from SCC is unreliable. The safest internal medicine framing is: a rapidly growing crateriform keratin-filled nodule is SCC until proven otherwise.

Feature	Keratoacanthoma / SCC-like lesion implication
Growth rate	Rapid enlargement over weeks is typical and concerning.
Morphology	Dome-shaped nodule with central keratin crater/plug.
Risk sites	Sun-exposed face, forearms, dorsal hands; higher concern in immunosuppression.
Management	Biopsy or excision rather than observation when diagnosis is uncertain; dermatology referral is appropriate.

6. Dangerous Mimics and Referral Triggers

Common skin disorders are common, but the clinician must recognize when the pattern is not behaving like the presumed diagnosis. A wart that grows rapidly, a tinea-like plaque that fails antifungals, a zoster-like eruption crossing many dermatomes, a scabies outbreak in a nursing facility, or an AK-like lesion that bleeds all require escalation. Dermatology referral is a safety mechanism for dermoscopy, biopsy, procedural therapy, and management in immunocompromised hosts.

Finding	Concern	Action
Rapidly growing keratotic nodule with central crater	Keratoacanthoma vs SCC	Biopsy/excision or urgent dermatology.
Nonhealing ulcer or bleeding lesion	SCC, BCC, melanoma, infection, vasculitis	Biopsy and evaluate infection/vascular context.
Dermatomal vesicles involving eye/forehead/nose	Herpes zoster ophthalmicus	Start antiviral therapy and urgent ophthalmology.
Diffuse vesicular or disseminated zoster eruption	Immunocompromised host or disseminated VZV	IV antiviral consideration, infection control, immune evaluation.
Widespread punched-out erosions in eczema patient	Eczema herpeticum	Urgent systemic antivirals; assess eyes.
Persistent generalized severe nocturnal pruritus with contacts affected	Scabies outbreak	Treat patient and contacts; environmental control.
Thick crusted plaques in institutionalized/immunosuppressed patient	Crusted scabies	Urgent treatment plus outbreak precautions.
Scaly annular rash worsened by steroid	Tinea incognito	KOH/culture and antifungal therapy; stop steroid monotherapy.
Pigmented lesion with asymmetry, border irregularity, color variation, diameter/evolution	Melanoma	Urgent biopsy/dermatology; do not treat as SK without confidence.

7. Integrated Diagnostic Algorithms

7.1 Verrucous Papule Algorithm

A verrucous papule on the hand or foot is usually a wart, but first examine context. If the lesion interrupts skin lines, has thrombosed capillaries, and is in a young patient, wart is likely. If pressure pain is vertical and skin lines continue through the lesion, callus or corn is more likely. If the lesion is solitary, rapidly enlarging, ulcerated, bleeding, or in an older or immunosuppressed patient, biopsy is preferred before destructive wart therapy. For genital verrucous lesions, manage as condyloma while assessing STI risk, pregnancy, immunosuppression, vaccination, and cancer screening needs.

7.2 Vesicular Rash Algorithm

For vesicles, ask: grouped or dermatomal; painful or itchy; localized or disseminated; mucosal or ocular involvement; immune status. Grouped recurrent vesicles suggest HSV. Unilateral dermatomal vesicles with pain suggest zoster. Widespread monomorphic punched-out erosions in eczema suggest eczema herpeticum. Vesicles with fever, hypotension, mucosal sloughing, or widespread skin pain suggest severe drug reaction or toxin-mediated disease rather than simple viral infection. PCR from lesion base helps when morphology is atypical.

7.3 Scaly Rash Algorithm

For scale, check distribution and perform KOH when uncertain. Annular plaques with central clearing suggest tinea corporis. Interdigital maceration suggests tinea pedis. Scalp alopecia with scale in a child suggests tinea capitis and needs oral therapy. Fine dyspigmented trunk scale suggests tinea versicolor. Greasy scale in seborrheic areas suggests seborrheic dermatitis. Thick silvery extensor scale suggests psoriasis. Gritty sun-exposed scale in an older adult suggests actinic keratosis. Failure to respond should trigger KOH, culture, biopsy, or referral.

7.4 Pruritus With Papules Algorithm

Nocturnal severe itch with papules in finger webs, wrists, waistline, genitalia, or household contacts suggests scabies. Scalp itch with nits or live insects suggests head lice. Flea or bedbug bites often appear in exposed areas and may cluster but do not cause burrows. Urticaria creates transient wheals that move and resolve within 24 hours at each site. Eczema is chronic and eczematous rather than burrow-like. Persistent generalized itch without primary lesions should prompt systemic evaluation rather than repeated empiric scabies treatment.

8. Treatment Pearls and Medication Safety

Therapy	Main uses	Safety / counseling
Salicylic acid	Common/plantar warts	Requires weeks of adherence; protect surrounding skin; avoid aggressive use in neuropathy or poor circulation without guidance.
Cryotherapy	Warts, AK, irritated SK	Pain, blistering, dyspigmentation; repeated treatments may be needed; caution on digits and in darker skin.

Therapy	Main uses	Safety / counseling
Imiquimod	Genital warts, selected AK/superficial skin cancer regimens	Local inflammation is expected; pregnancy/site-specific considerations; adherence matters.
Acyclovir/valacyclovir/famciclovir	HSV and VZV	Renal dosing; start early; IV acyclovir for severe/disseminated disease.
Topical terbinafine/azoles	Localized dermatophyte infection	Continue long enough; treat reservoirs; do not combine indiscriminately with steroids.
Oral terbinafine/griseofulvin/azoles	Tinea capitis, onychomycosis, extensive dermatophyte disease	Check interactions, liver disease context, pregnancy considerations; confirm nail diagnosis before long courses.
Permethrin 5%	Scabies	Apply thoroughly and repeat; treat contacts; post-scabetic itch is not always failure.
Oral ivermectin	Scabies alternative/outbreaks/crusted scabies regimens	Not FDA-approved for scabies in the US; avoid or use specialist guidance in pregnancy and small children; repeat dosing required.
5-FU/imiquimod/tirbanibulin/diclofenac/PDT	Field therapy for AK	Inflammatory reaction expected; counsel before treatment; avoid 5-FU in pregnancy.

9. Graph-RAG Concept Map

This compact concept map is intended for future node-edge extraction. It emphasizes entities, diagnostic clues, relationships, and treatment edges.

Concept node	High-yield attributes	Important edges
HPV wart	Verrucous hyperkeratotic papule; common, flat, plantar, anogenital subtypes.	HPV 6/11 -> condyloma; HPV 16/18 -> cancer risk; salicylic acid/cryotherapy -> lesion destruction.
Molluscum contagiosum	Poxvirus; umbilicated papules; children, sexual transmission, immunosuppression.	Impaired cellular immunity -> extensive molluscum; curettage/cryotherapy/cantharidin -> lesion clearance.
Herpes zoster	VZV reactivation; unilateral dermatomal painful vesicles.	Age/immunosuppression -> zoster risk; V1 involvement -> ophthalmology; antivirals within 72h -> decreased acute severity.
Dermatophyte infection	Tinea by site; KOH septate hyphae; keratin infection.	Tinea capitis/nails -> oral therapy; steroid exposure -> tinea incognito; moisture/occlusion -> tinea pedis/cruris.
Tinea versicolor	Malassezia; dyspigmented fine scale; KOH spaghetti and meatballs.	Heat/humidity/oily skin -> overgrowth; selenium sulfide/azoles -> treatment; pigment lag -> persistent dyspigmentation after cure.
Scabies	Sarcoptes; nocturnal pruritus, burrows, contacts affected.	Permethrin/ivermectin -> mite eradication; untreated contacts -> reinfestation; crusted scabies -> outbreak risk.
Actinic keratosis	Premalignant UV keratinocyte lesion; gritty sun-exposed scale.	UV damage -> field cancerization; hypertrophy/induration/bleeding -> biopsy for SCC; cryotherapy/field therapy -> treatment.
Seborrheic keratosis	Benign stuck-on waxy plaque; no malignant potential.	Diagnostic uncertainty -> biopsy; irritation/cosmesis -> cryotherapy/curettage.
Keratoacanthoma	Rapid crateriform keratin-filled nodule.	Clinical overlap with SCC -> excision/biopsy; immunosuppression/UV -> higher SCC concern.

10. Review Cases

Question: A 12-year-old has patchy alopecia with scale and posterior cervical lymph nodes. KOH is positive. What is the key treatment principle?

Answer: Tinea capitis requires oral antifungal therapy because topical agents do not adequately penetrate infected hair shafts. Adjunct antifungal shampoo can reduce shedding but is not sufficient alone.

Question: A 74-year-old has a gritty pink papule on the dorsal hand that becomes tender and thick despite cryotherapy. What should happen next?

Answer: Biopsy to exclude SCC. Tenderness, hypertrophy, induration, bleeding, ulceration, rapid growth, or treatment failure converts an AK-like lesion into a biopsy problem.

Question: A patient with severe nocturnal itch has papules on finger webs and wrists; his roommate also itches. What management step prevents failure?

Answer: Treat close contacts simultaneously and decontaminate bedding/clothing. Persistent itch after treatment can be post-scabetic and does not always mean active infestation.

Question: A unilateral painful vesicular rash appears on the forehead and nose. What is the emergency?

Answer: Herpes zoster ophthalmicus. Start systemic antiviral therapy and obtain urgent ophthalmology evaluation.

Question: A scaly annular plaque worsened after topical steroid. What diagnosis should rise?

Answer: Tinea incognito. Stop steroid monotherapy, perform KOH/culture if needed, and treat with antifungal therapy.

Selected References

1. American Academy of Dermatology. Clinical guidelines and pathways for actinic keratosis and other dermatology conditions. Accessed 2026.
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3. Centers for Disease Control and Prevention. Clinical Care of Scabies. Updated 2023.
4. Centers for Disease Control and Prevention. Sexually Transmitted Infections Treatment Guidelines: Genital Herpes and Ectoparasitic Infections.
5. Merck Manual Professional Edition. Dermatologic disorders: viral skin infections, fungal skin infections, scabies, and keratinocyte lesions.
6. Uploaded Medvale dermatology source segment derm2.pdf, pages 1-15, used as the local coverage guide for warts, molluscum, zoster, dermatophytes, tinea versicolor, scabies, actinic keratosis, seborrheic keratosis, and keratoacanthoma.

Appendix A. Bacterial Skin Infection Bridge

The uploaded source directs bacterial infections back to the infectious disease chapter, but a dermatology-focused review benefits from a compact bedside bridge. Bacterial skin disease is organized by depth and morphology: superficial epidermal infection, follicular infection, abscess/purulence, nonpurulent cellulitis, toxin-mediated disease, and necrotizing infection. Internal medicine exams often test whether the clinician recognizes purulence, systemic toxicity, necrotizing features, and host risk rather than memorizing every antibiotic.

Syndrome	Morphology and distribution	Common pathogens	Diagnostic approach	Management principle
Impetigo	Honey-colored crusted erosions, usually face/extremities; bullous form has flaccid bullae.	Staphylococcus aureus; Streptococcus pyogenes; bullous form is toxin-producing S. aureus.	Clinical; culture if outbreak, MRSA concern, or failure.	Topical mupirocin/retapamulin for limited disease; oral therapy for extensive disease; hygiene and school/contact precautions.
Ecthyma	Deeper punched-out ulcer with adherent crust, often legs.	Strep pyogenes and/or S. aureus.	Clinical; culture if severe or refractory.	Oral antibiotics because infection extends into dermis.
Folliculitis	Follicular pustules or papules in shaved/occluded areas; hot tub folliculitis after whirlpool exposure.	S. aureus; Pseudomonas for hot tub folliculitis; Malassezia can mimic.	Clinical; culture if recurrent or severe.	Antiseptic washes, topical/oral anti-staphylococcal therapy when needed; remove friction/occlusion; hot tub form often self-limited.
Furuncle/carbuncle	Tender follicular abscess; carbuncle is coalescent deeper abscess with multiple drainage points.	S. aureus including MRSA depending on setting.	Look for fluctuance; culture purulent drainage.	Incision and drainage is central; antibiotics when systemic signs, extensive disease, immunosuppression, recurrent disease, or difficult site.
Cellulitis	Diffuse warmth, erythema, edema, tenderness; usually unilateral lower extremity.	Nonpurulent: beta-hemolytic streptococci; purulent: S. aureus/MRSA more likely.	Clinical; evaluate portals of entry; ultrasound if abscess uncertain.	Nonpurulent cellulitis targets streptococci; purulence shifts toward drainage and MRSA-active therapy when indicated.
Erysipelas	Sharply demarcated raised erythematous plaque, often face or legs; lymphatic involvement.	Strep pyogenes most classic.	Clinical.	Streptococcal therapy; address edema, tinea pedis, venous disease, and recurrent risk factors.
Necrotizing soft tissue infection	Pain out of proportion, rapidly progressive edema/erythema, bullae, skin anesthesia, crepitus, toxicity, shock.	Polymicrobial, group A strep, clostridial species, Vibrio/Aeromonas in exposure contexts.	Do not delay surgery for imaging if high suspicion; labs may support but cannot exclude.	Emergent surgical exploration/debridement plus broad antibiotics and hemodynamic support.

Purulence rule: If there is a drainable abscess, incision and drainage is the therapeutic core. Antibiotics are adjunctive when there are systemic signs, surrounding cellulitis, immunosuppression, extremes of age, multiple lesions, recurrent disease, difficult-to-drain sites, or failure of drainage alone.

A.1 Mimics of Cellulitis

Cellulitis is overdiagnosed because many inflammatory, vascular, and edematous disorders cause red legs. True cellulitis is usually unilateral, warm, tender, and acute; bilateral chronic lower-extremity erythema should raise suspicion for venous stasis dermatitis, lipodermatosclerosis, lymphedema, contact dermatitis, gout, DVT, or inflammatory dermatoses. Fever, leukocytosis, lymphangitis, portal of entry, and response to antibiotics support infection, but absence of fever does not exclude cellulitis.

Mimic	Clues favoring mimic over cellulitis	Useful next step
Venous stasis dermatitis	Bilateral, chronic, pruritic, scaling, hemosiderin, edema; less tender than cellulitis.	Compression/edema management, topical steroids for dermatitis, evaluate venous disease.
Lipodermatosclerosis	Chronic induration, inverted champagne-bottle legs, hyperpigmentation; may flare painfully.	Venous management; avoid repeated antibiotics unless infection signs.
DVT	Unilateral swelling/pain, risk factors; erythema may be mild.	Duplex ultrasound when pretest probability supports.
Gout/pseudogout	Acute monoarticular pain, erythema centered over joint.	Joint exam, aspiration if septic arthritis possible.
Contact dermatitis	Pruritic, sharply patterned/geometric eruption; exposure history.	Remove trigger; topical steroids; infection only if secondary features.
Necrotizing infection	Severe pain, toxicity, rapid progression, bullae, crepitus, anesthesia.	Immediate surgical evaluation.

Appendix B. Immunocompromised Host and Infection Control Patterns

Immunocompromised patients can have atypical, extensive, disseminated, or treatment-resistant skin disease. The same morphology can mean a different risk level. Extensive molluscum may suggest impaired cellular immunity. Multidermatomal or disseminated zoster can reflect immunosuppression and requires infection control attention. Crusted scabies has enormous mite burden and can cause institutional outbreaks even when itch is minimal. Dermatophyte infection can be more extensive and steroid-modified. Nonhealing keratotic lesions are more concerning for SCC, especially in solid-organ transplant recipients and patients receiving chronic immunosuppression.

Pattern	Why immunosuppression matters	Escalation
Extensive molluscum	Large, numerous, facial, or refractory lesions may indicate impaired cellular immunity.	Assess HIV risk/immune status; dermatology if extensive.

Pattern	Why immunosuppression matters	Escalation
Disseminated zoster	Risk of visceral involvement and transmission; may require IV therapy.	Airborne/contact precautions in many settings; IV acyclovir and immune evaluation.
Crusted scabies	Minimal itch but massive mite load; highly contagious.	Treat with oral plus topical regimens; contact tracing, environmental control, outbreak management.
Refractory dermatophyte	More extensive disease; atypical morphology after steroids.	KOH/culture, oral therapy when appropriate, review immunosuppressive medications.
Keratotic lesion in transplant patient	Higher SCC incidence and aggressiveness.	Lower threshold for biopsy and dermatology surveillance.

Appendix C. Rapid Test and Biopsy Selection

Clinical problem	Best initial test or action	Reasoning
Annular scaly plaque	KOH scraping from active border.	Highest-yield quick discriminator between tinea and eczema/psoriasis mimics.
Dyspigmented fine trunk scale	KOH scraping.	Tinea versicolor shows short hyphae/spores; pigment change can persist after cure.
Scalp scale with alopecia	KOH/culture; consider Wood lamp but do not rely on it.	Tinea capitis usually needs oral therapy; Wood lamp sensitivity depends on organism.
Nocturnal itch with burrows	Mineral oil scraping or dermoscopy if available; clinical treatment often reasonable.	Microscopy can show mites/eggs/scybala, but sensitivity is imperfect.
Fresh vesicles, atypical HSV/VZV, immunocompromised host	Viral PCR from lesion base.	PCR is more sensitive and specific than clinical impression when pattern is atypical.
Purulent abscess	Drainage and culture if moderate/severe/recurrent or MRSA concern.	Source control is therapeutic; culture guides recurrent or complicated disease.
Nonhealing, bleeding, indurated, rapidly growing keratotic lesion	Biopsy rather than repeated empiric treatment.	Rules out SCC, BCC, melanoma, atypical infection, or inflammatory mimics.
Pigmented lesion suspicious for melanoma	Excisional biopsy when feasible or dermatology-directed biopsy.	ABCDE/evolution features should not be destroyed with cryotherapy.

Appendix D. Patient Counseling Scripts

Condition	Counseling points
Warts	Benign HPV-related growths; many regress but treatment often takes multiple attempts. Do not pick or shave through lesions. Genital warts require STI context, HPV vaccination review, and routine cancer screening.
Molluscum	Often self-limited but contagious by direct contact. Avoid scratching, sharing towels, and uncovered contact sports. In adults with genital lesions, consider sexual transmission and STI testing.
Zoster	Start antivirals early. Keep lesions covered until crusted. Seek urgent care for eye involvement, facial weakness, confusion, widespread rash, or immunosuppression. Vaccination prevents future episodes in eligible groups.
Tinea	Use antifungals for the full course and keep area dry. Avoid steroid-only creams. Treat feet/nails if they are reservoirs for groin or body recurrence.
Scabies	Itching may continue after mites are dead. Treat close contacts on the same day. Wash/dry bedding and clothes or bag items that cannot be washed. Return for new burrows or ongoing spread.
Actinic keratosis	AK reflects sun damage and can be part of a field of precancerous change. Use sun protection and return for lesions that thicken, hurt, bleed, ulcerate, or grow.
Seborrheic keratosis	Benign stuck-on growths are common with aging. Treatment is optional unless irritated or uncertain. A changing dark lesion should be checked rather than assumed benign.